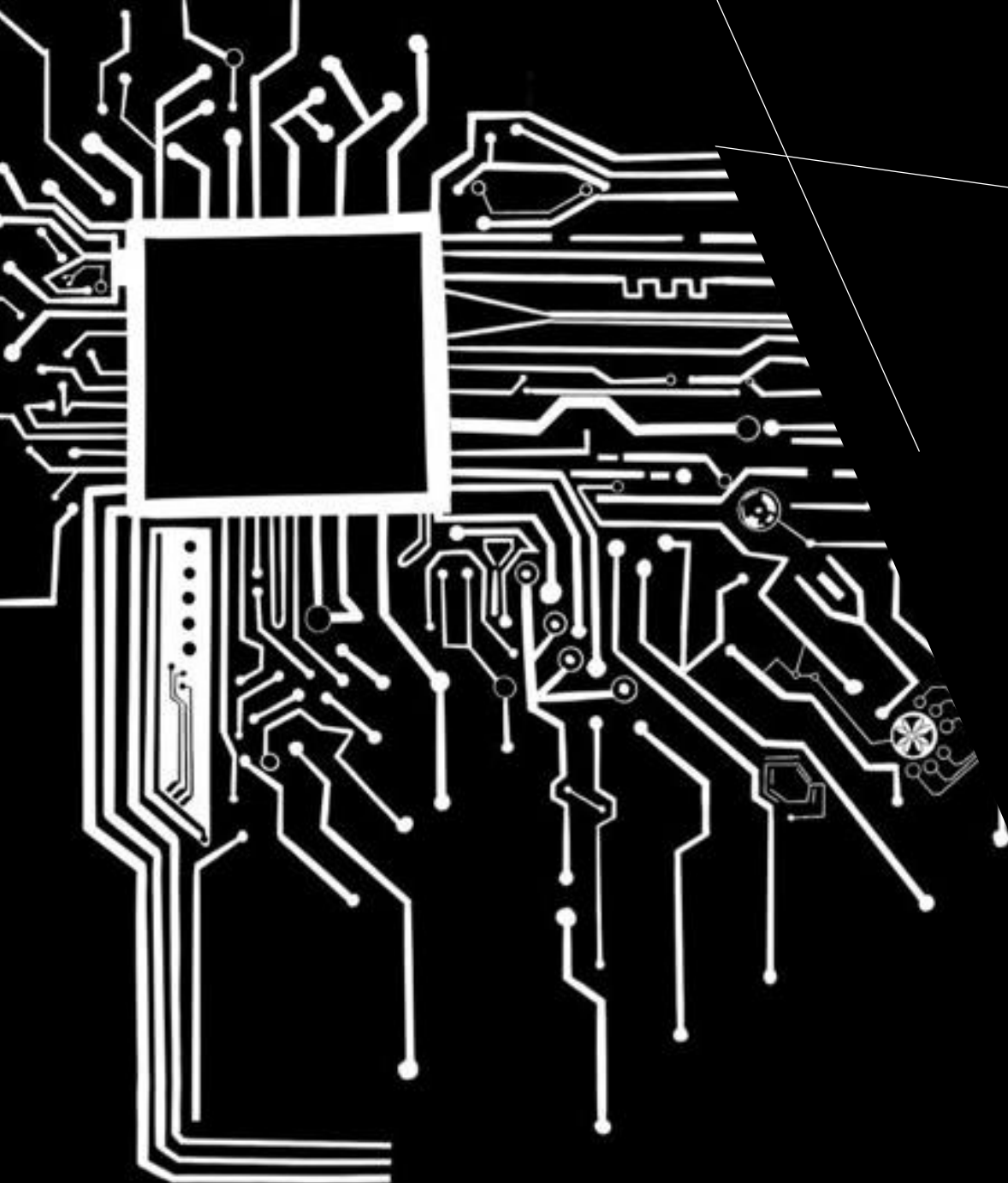


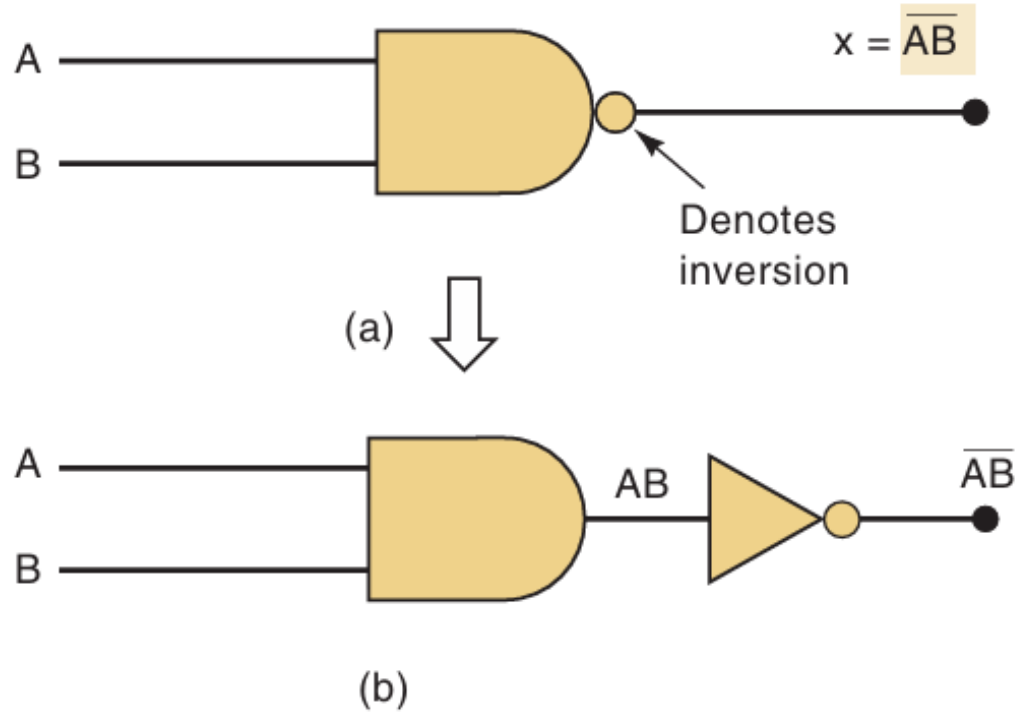


ICE 2343

DIGITAL LOGIC DESIGN

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NAND Gate



		AND		NAND	
A	B	AB		$\overline{AB}$	
0	0	0		1	
0	1	0		1	
1	0	0		1	
1	1	1		0	

(c)

Universality of NAND

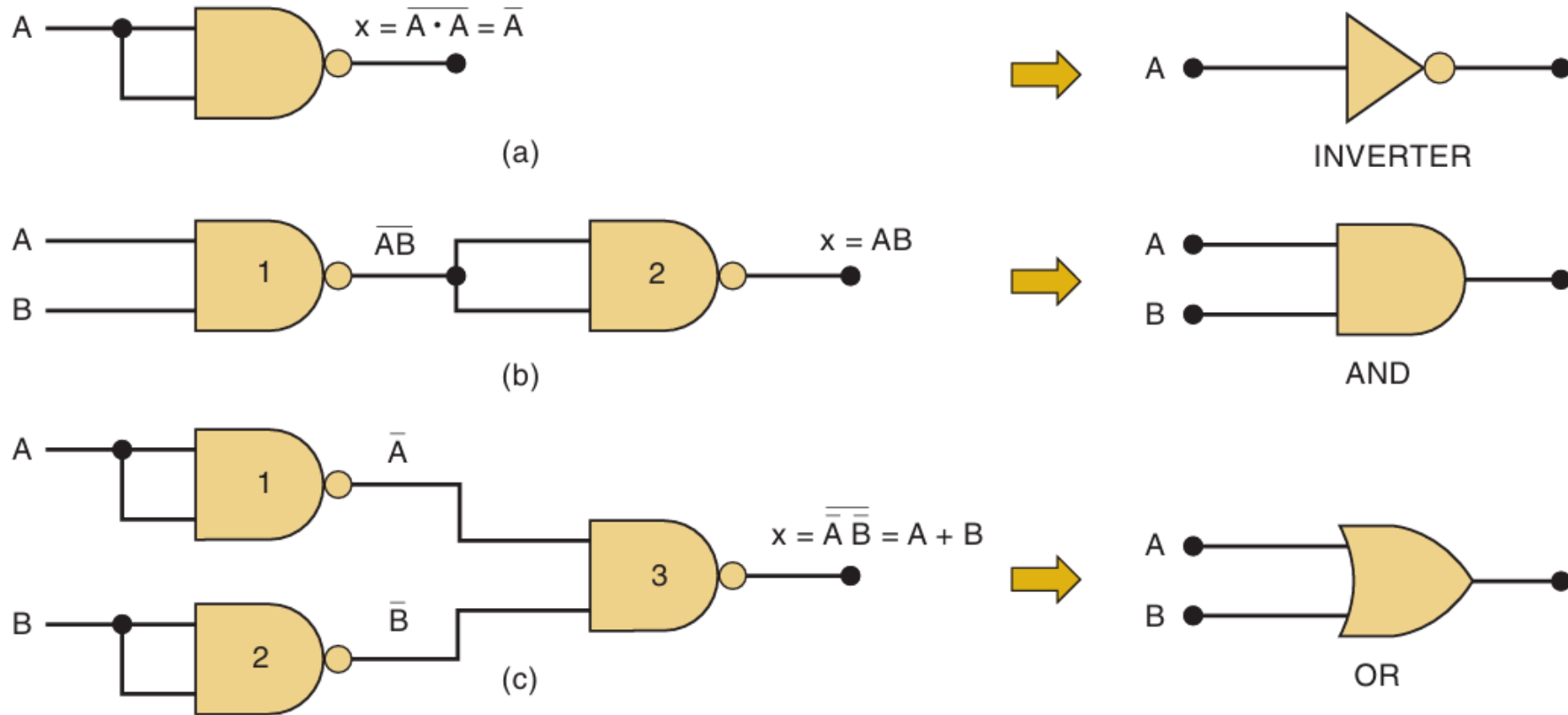
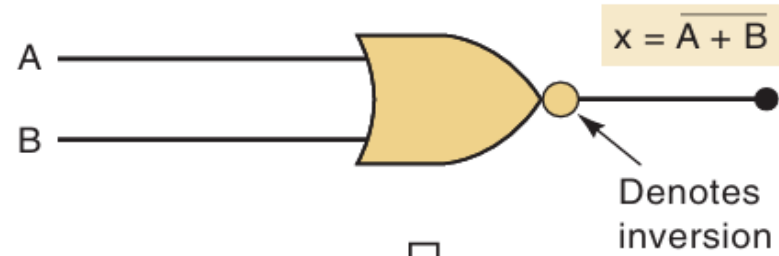
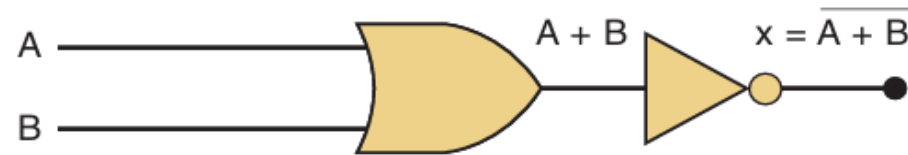


FIGURE 3-29 NAND gates can be used to implement any Boolean function.

NOR Gate



(a) ↓



(b)

		OR		NOR	
A	B	$A + B$		$\overline{A + B}$	
0	0	0		1	
0	1	1		0	
1	0	1		0	
1	1	1		0	

(c)

Universality of NOR

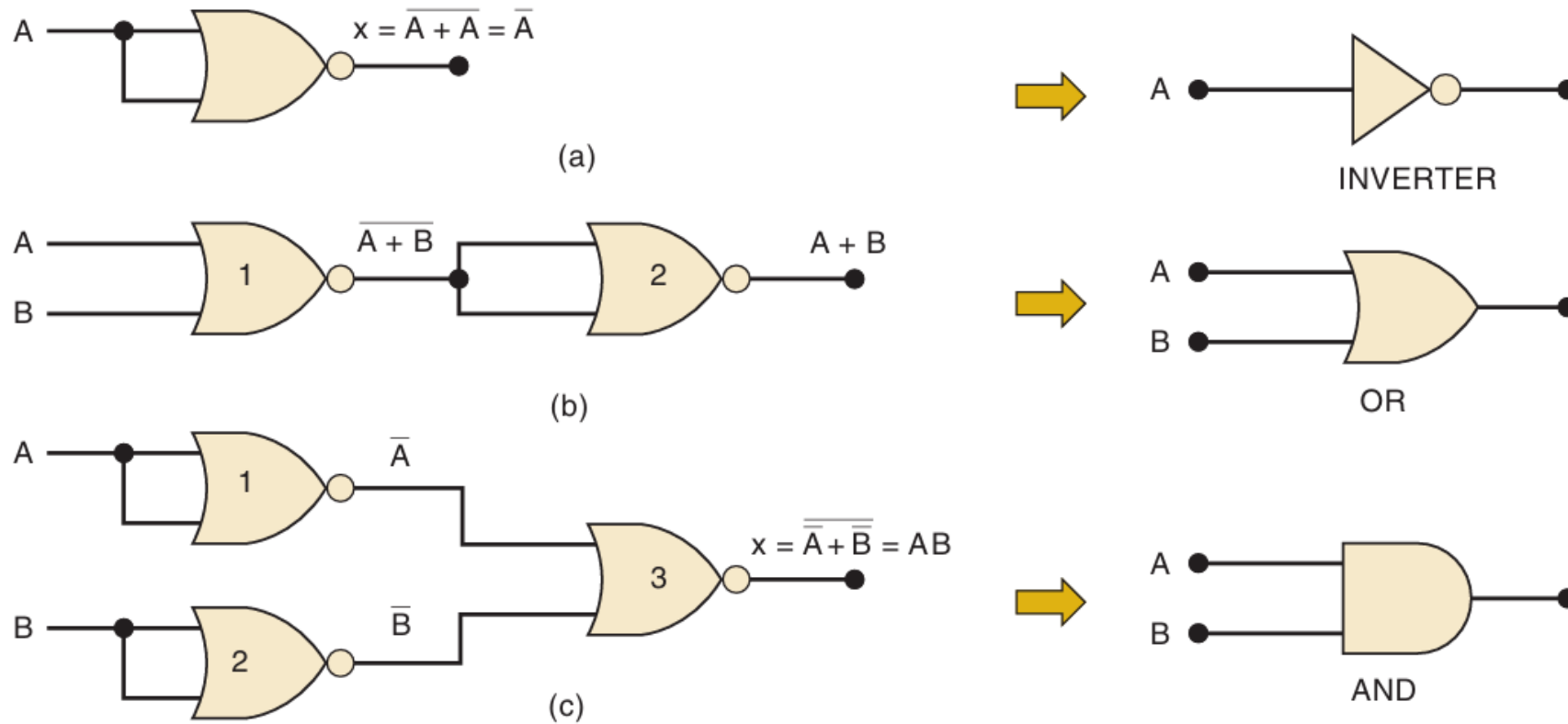
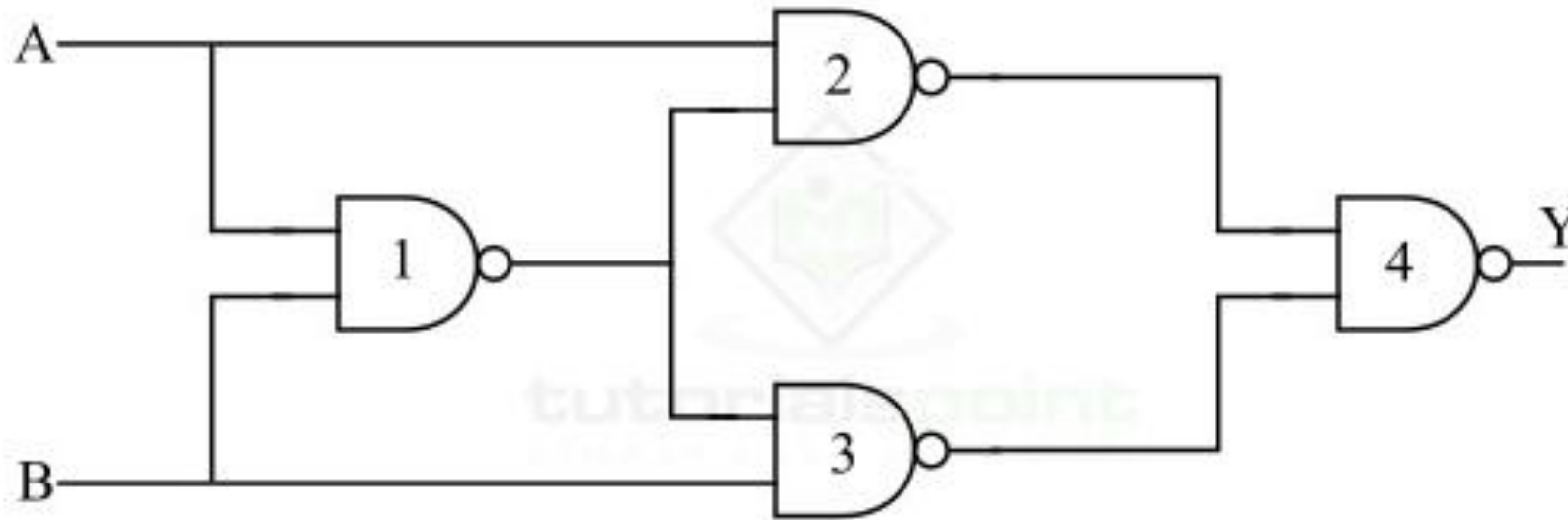


FIGURE 3-30 NOR gates can be used to implement any Boolean operation.

Universal Gates

Realize the following Boolean expression using NAND gate: $A'B+AB'$



This is basically XOR gate.

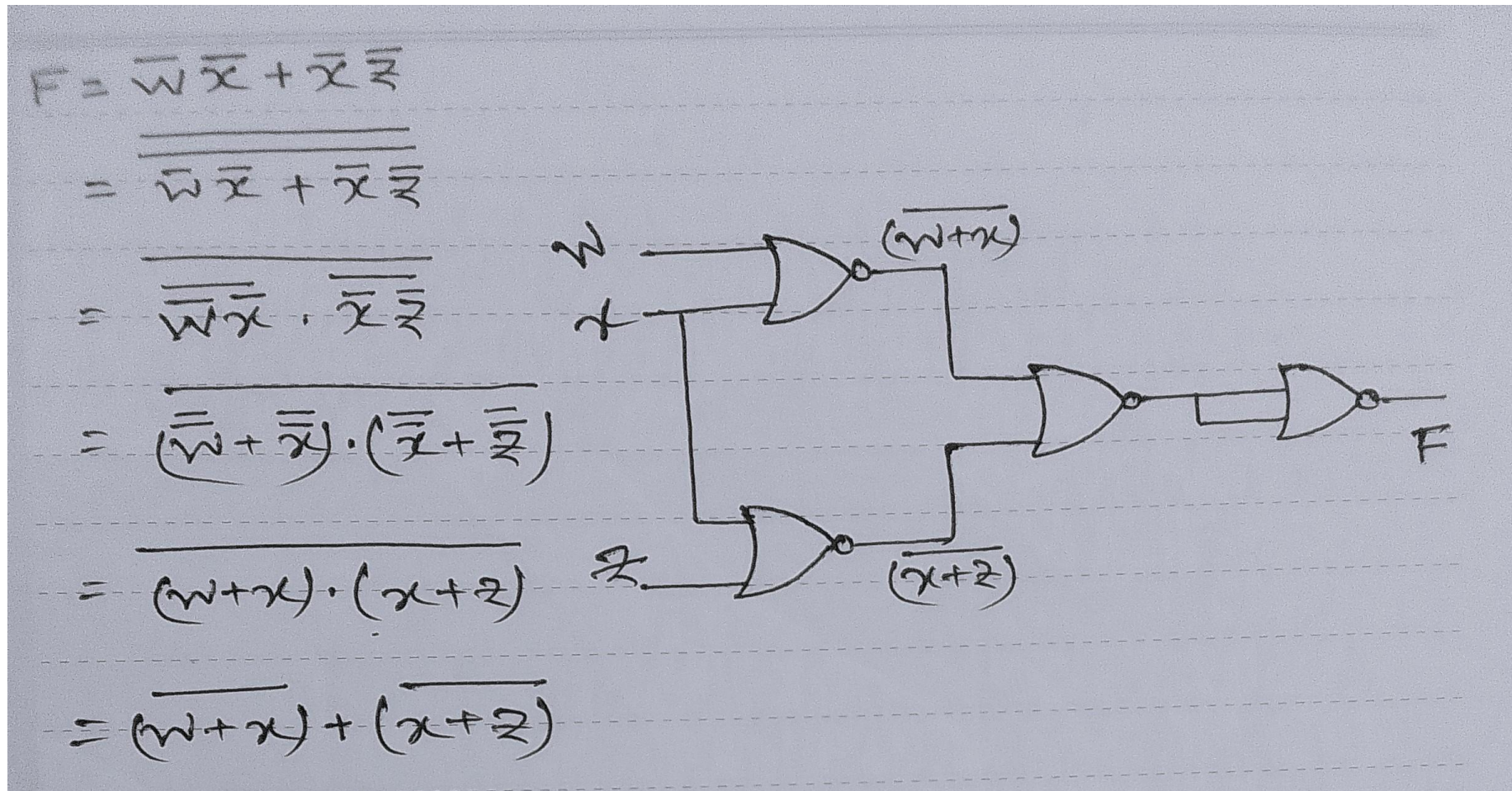
Now, implement the same expression by NOR gate.

Boolean Expression Using Universal Gates

- ❑ To implement a Boolean expression by universal gate, first take double complement of the given expression or equation.
- ❑ Apply de morgan's law to realize the expression using only product and complement of variable for NAND gate
- ❑ Apply de morgan's law to realize the expression using only sum and complement of variable for NOR gate
- ❑ According to final expression draw the circuit using required universal gate.

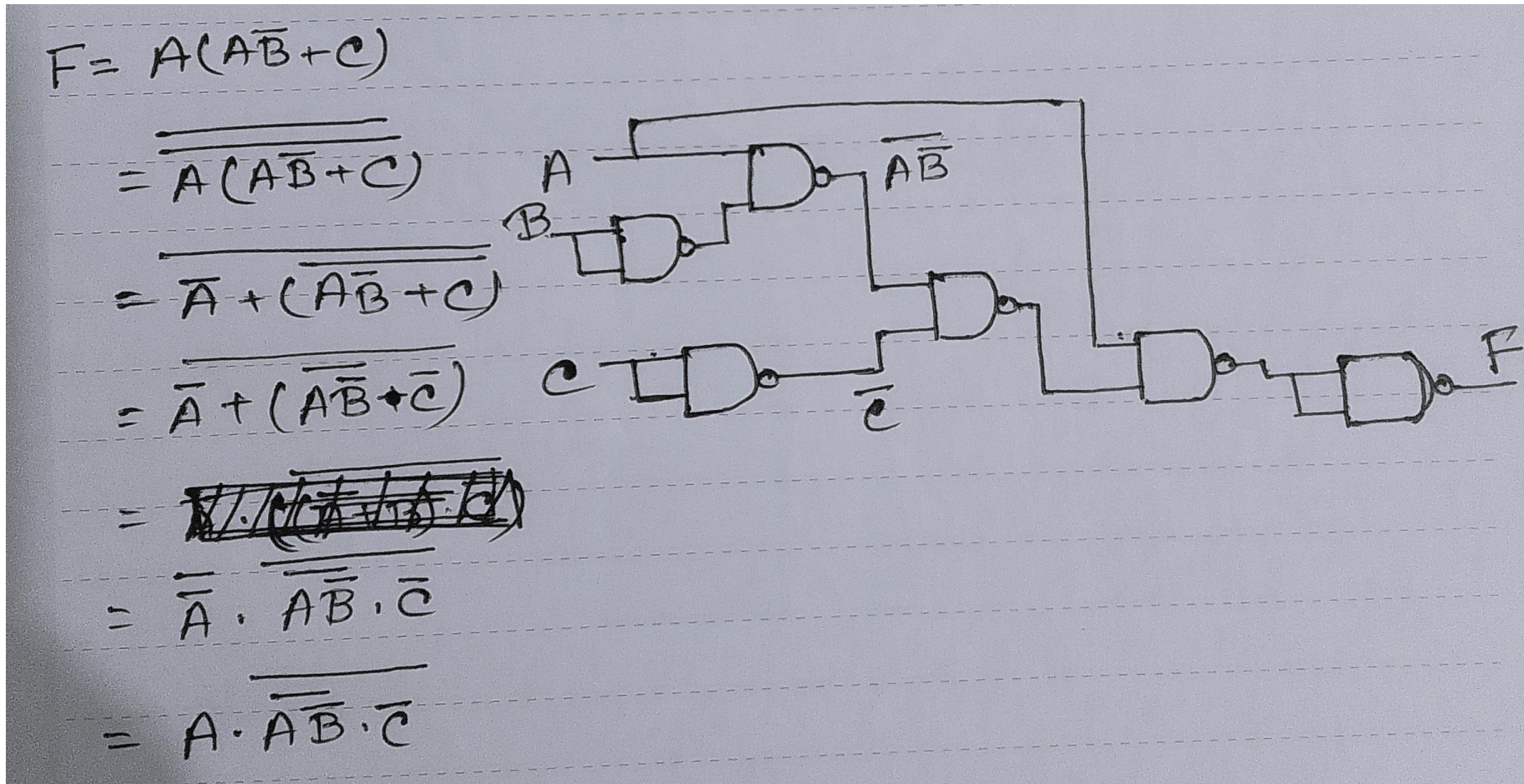
Universal Gates

Problem 1: Implement using NOR gate $F = w'x' + x'z'$



Universal Gates

Problem 2: Implement using NAND gate $F = A(\overline{AB} + C)$

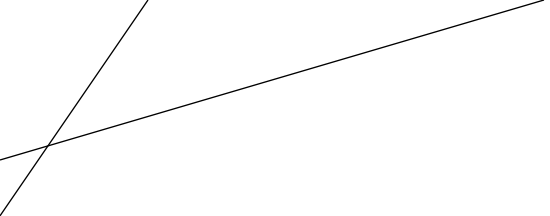


Practice

Implement Problem 1 using NAND gate and Problem 2 using NOR gate

Problem 3: $F = X'Y' + XY + Y'Z$ implement using NAND gate

Problem 4: $F = (A' + B')C$ implement using NOR gate



Reference

Digital Systems Principles and Applications by Ronald J.Tocci

A series of white, thin, overlapping geometric lines on a black background, forming various polygons and intersecting points, primarily located on the left side of the slide.

THANK YOU